

LEVERAGING EXPANDABLE MODULES, ORIGAMI TESSELLATIONS AND THE RE-USABLE HEAVY LIFT ROCKETS TO CREATE VAST PRESSURIZED ECOSYSTEMS IN THE LUNAR LAVA TUBES

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1. ABSTRACT :

The Starship and New Glenn rockets fairing size allow us to make expandable modules so large they could plug most lunar lava tubes. Plugging, sealing, and coating a lava tube would permit pressurization, providing habitable space of significant size.

2. BACKGROUND AND INTRODUCTION :

There is a growing consensus in the space exploration community that expandable modules are the most efficient means of providing living and working space outside the confines of our atmosphere.

Fig 2.0 Expandable habitat in Orbit



Nearly all deployable structures, whether rigid or flexible, can benefit from origami folding techniques for tight packing and easy unfolding, be it airbags, disaster relief shelters or even plumbing.[1] Some radial origami tessellations have already been used to facilitate the unfurling of satellite hardware such as solar cells and sails.[2] Many such techniques remain unexplored in scope and applications.

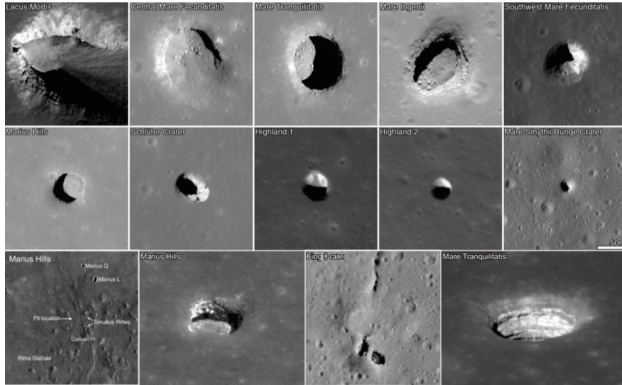
Fig 2.1 Self-deploying origami structure



A 9m diameter fairing such as Starship's would allow for an expanded habitat diameter of 50ft to reach orbit[3]. Beyond providing considerable legroom to astronauts in deep space, such structures could be buried under the surface of rocky worlds to offer protection against radiation and micrometeorites. Advocates for such projects have suggested lava tubes on the Moon and Mars as ideal sites for early settlement in order to minimize excavation and burial of the structures which are essential for protection against radiation.

Since 2008 the Lunar Reconnaissance Orbiter has found hundreds of pits and caves of various depth, size and geometry pockmarking the Moon, some of which are plausible lava tube entrances. Geologists have speculated that lunar rilles could result from the collapse of such cave networks[4], implying the potential existence of vast, sprawling cavities beneath our cosmic neighbor's surface.

Fig 2.2 Lunar lava tube entrances



It could theoretically be possible to use a large expandable module with deployable skirts in combination with sealing agents to enclose and even pressurize such caves.

3. METHODS AND CHALLENGES :

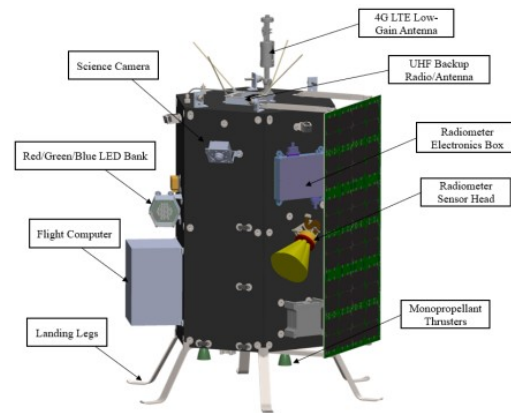
This project would face significant challenges, the least of which being the insofar limited knowledge of what the depth of those tubes contain, steep denivelation, abrasive lunar dust, uneven tunnel opening shape, and the unknown structural integrity of such formations.

This section of the document will provide an overview of the designs and engineering that could overcome these challenges and allow for hundreds to live inside the Moon by the end of the next decade.

3.1 ROBOTIC SURVEY:

In order to determine if one of the lava tube sites discovered by the LRO could become inhabited, a thorough radar and X-ray survey of its content and structural integrity must be performed. A swarm of drones capable of jumps and/or controlled airless flight should form the core of this phase's surface assets.

Fig3.0 Micro Nova Hopper



Those would be deorbited from a basecamp in polar orbit, along with the equipment required to relay power and communication through beam or cable into the cave.

3.2 SITE PREPARATION:

Once a site of sufficient depth and structural integrity for occupation has been found, a swarm of sinter-bots, dozer-bots and climbing chipper-bots are brought down from the orbital basecamp to prepare the landing site and fashion the lava tube opening into a geometry compatible with the inflatable module. Sinter-bots compact the regolith and sinter it to make proper landing pads in order to avoid kicking up debris with larger landers, a problem discovered during Apollo 12 when the manned lunar lander damaged nearby Surveyor 3 [5].

The chipper bots use jackhammers to sculpt the tunnel opening and chip away the rocks that are too large to be excavated by the dozers-bots. Dozer-bots also move the regolith to form ramps if tunnel denivelation is too steep.

Fig3.1:Sinter-bots compacting and sintering the regolith to build a landing site.

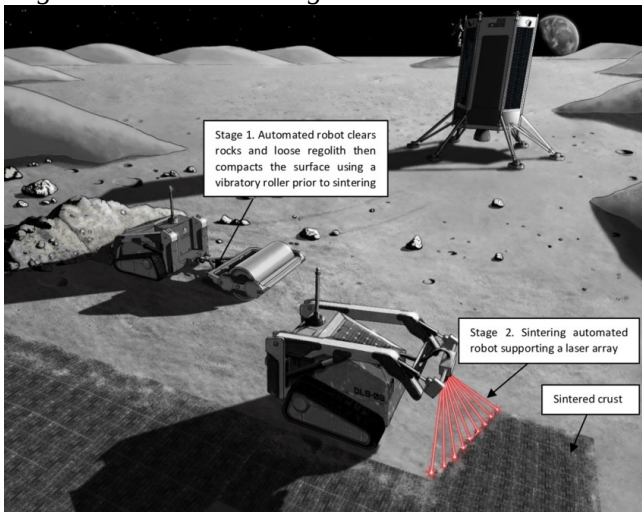


Fig 3.2 Dozer bot by Offworld Industries (right); Boston Dynamics Spot robot with microspine gripper feet and jackhammer arm attachment used as Chipper bot (left)



Once a landing pad capable of supporting heavy landers is ready, a human habitation module and a team of astronauts are brought to the surface to supervise the operation and perform necessary repairs to the robot fleet. The biological waste they produce is stored for atmospheric nitrogen production in the later phases of this project, and to kickstart a self-expanding carbon loop via PLA-producing bioreactors[6].

The abrasive nature of lunar regolith presents a significant challenge to the full use of the enclosed space. Sinter-bots are repurposed to melt the surfaces of the cave, which are then coated with locally biosynthesized PLA by plastic extruders mounted on repurposed chipper-bots.

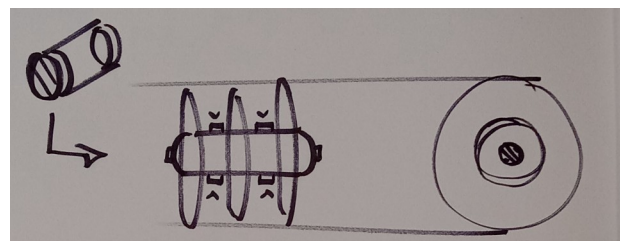
3.3 INFLATABLE MODULE TAXI:

Once the lava tube entrance has been cleared of debris, the large inflatable module is deorbited and brought into position at the mouth of the cave via lander and rover. Additional power units such as RTGs, solar and hydrogen cells are brought to the site to fuel the operation; chipper and dozer-bots are repurposed for ice mining either within the cave or in a nearby permanently shadowed area. ISRU capabilities will be crucial for the following steps of the project.

3.4 INFLATABLE MODULE DEPLOYMENT :

The large inflatable module consists of a cylindrical main body and at least two annular skirts that are creased in a concentric pattern for close packing. It has a hatch at both domed extremities of the cylinder; it is deployed along the length of the lava tube with the use of pressurized canisters.

Fig3.4 Large Inflatable module

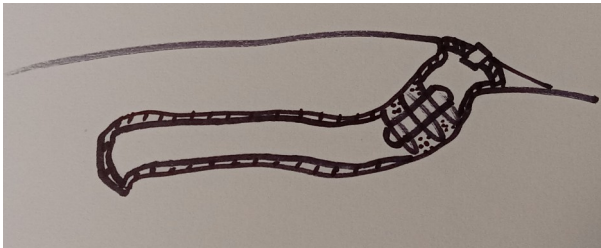


Once the flaps of the skirts brace against the walls of the tunnel a hardening polymer foam is injected between those flaps and the cylinder to seal the tunnel opening. Sinter bots enclose the tunnel's entrance with a dome and have it capped with an airlock.

3.5 LAVA TUBE PRESSURIZATION :

Once the lava tube is sealed, it can be pressurized with mined oxygen, carbon dioxide and imported nitrogen. Solar-powered electric heaters and lamps are installed to provide warmth and light. Batteries to last through the lunar night can be made from ISRU elements [7]

Fig3.5 : Cross section of the modified lava tube with inflatable module, PLA coating and outside sintered regolith dome



4. OPERATION:

Once a warm environment has been established under the surface of the moon, it can be expanded and an arcology can be built inside.

4.1 EXPANSION:

The end part of the tunnel is carved non-stop by crews of robots and astronauts to expand the inside of the habitat. The walls are coated with more PLA as the tunnel gets deeper and plastic film partitions ensure limited airborne regolith exposure for the rest of the cave. The excavated regolith is either used to make more soil, or it is compressed and sintered/cast into biocrete bricks, most of which are brought to the surface every time the astronauts have to go out. This reduces atmospheric loss of every airlock cycle. The bricks can be used or sold to form structures on the surface of the moon.

4.2 ARCOLOGY:

Biocrete cuboctahedric construction nodes, ISRU derived steel cables and imported/locally grown high strength bamboo cultivars are used for structural framing; imported/locally made plywood form floor sections.

The bottom level of the arcology is occupied by a Self Organizing Wetland Bioreactor that performs extraction and separation for repurposing of the various elements in the water cycle of the habitat. Veins of free-form agriculture modules made from 3D-printed PLA rise from the SOWB and span across the entire structure, some of which are white-ish clear with a stream of green algae coursing under its surface, others are dark brown, holding nourishing soil for the plants growing from them. The entire ecosystem maintains homeostasis through a predictive AI using a modular neural network. [8]

Aquaculture of shrimp and tilapia, sweet potatoes and arconaut-selected plants and animals provide food, medicine and atmosphere. [9] Mealworms and algae can provide feed for the animals.

No traditional Earth-based farming technique is off the table inside a decent gravity well; partial rotational gravity can always be explored if adaptation is a challenge.

This environment is designed to create an overabundance of fresh goods that can either be exported elsewhere on the surface (or orbit) of the moon for a massive profit, or traded for logistical benefits.

5. DISCUSSION:

The arcology design is based on ArcologyX's model

Reference:

[1]:<https://phys.org/news/2015-09-paper-tubes-stiff-origami.html>

[2]:<https://www.jpl.nasa.gov/news/solar-power-origami-style>

[3]:<https://www.youtube.com/watch?v=C3bd4fdY3DY&t=1666s>

[4]:<https://volcano.oregonstate.edu/sinuuous-rilles>

[5]: <https://www.nasa.gov/wp-content/uploads/static/history/alsj/a12/AnalysSurvIIIMtrial.pdf>
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[6] https://isruinfo.com/public/docs/srr20_ptmss/8-6%20Architecture%20for%20a%20Farm%20in%20a%20Moon%20Village%20with%20In-Situ%20Materials-Muscatello.zip

[7]:<https://www.youtube.com/watch?v=2TigGaT2C-o&t=1015s>

[8]:<https://www.seasteading.org/biodiverse-habitats-with-arcx/>

[9]:<https://youtu.be/QMHpscIm5Wo?t=479>

Fig2.0 Expandable Habitat

<https://www.rocketstem.org/2014/03/12/city-in-the-sky-building-a-cheaper-space-station-with-bigelow-modules/>

Fig2.1 Self Deploying Origami Structure

<https://www.youtube.com/watch?v=1YSEDUwd84A>

Fig2.2 Lunar Lava Tube Entrance

<https://www.universetoday.com/163371/chinese-astronauts-may-build-a-base-inside-a-lunar-lava-tube/>

Fig3.0 Micro Nova Hopper

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Fig3.1 Sinter-bot

https://www.researchgate.net/figure/Robots-direct-laser-sintering-the-lunar-surface-Two-construction-robots-are-shown_fig1_363586646

Fig 3.2 Chipper bot and dozer bot

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<https://www.offworld.ai/post/offworld-is-now-taking-development-contracts-for-new-ai-powered-swarm-robots>